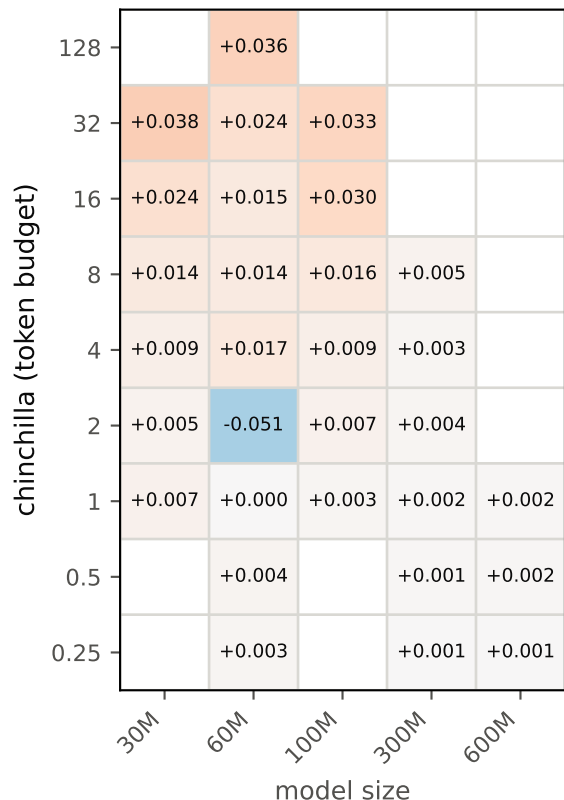
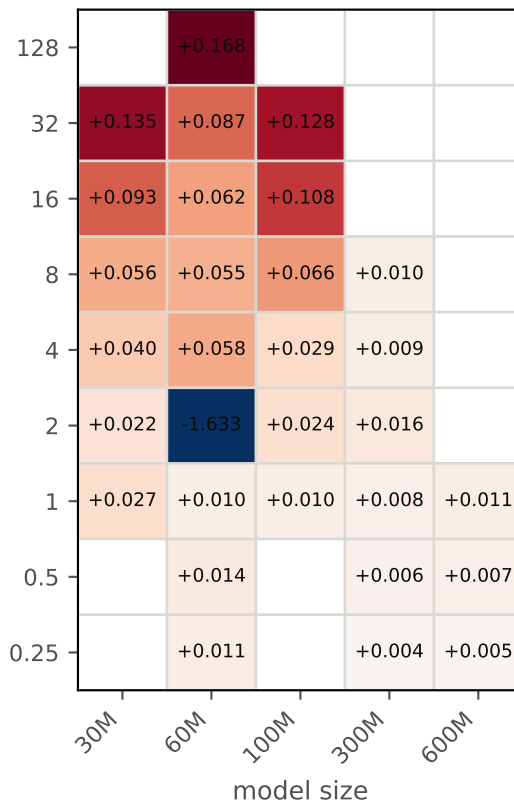


Relative degradation under Gaussian weight perturbation: muon vs adamw ($\gamma \leq 0.02$)

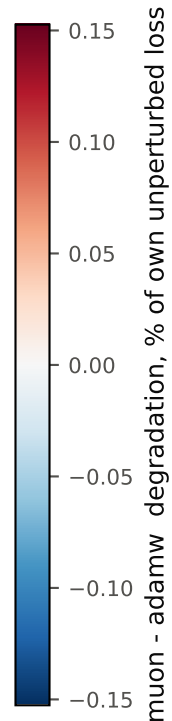
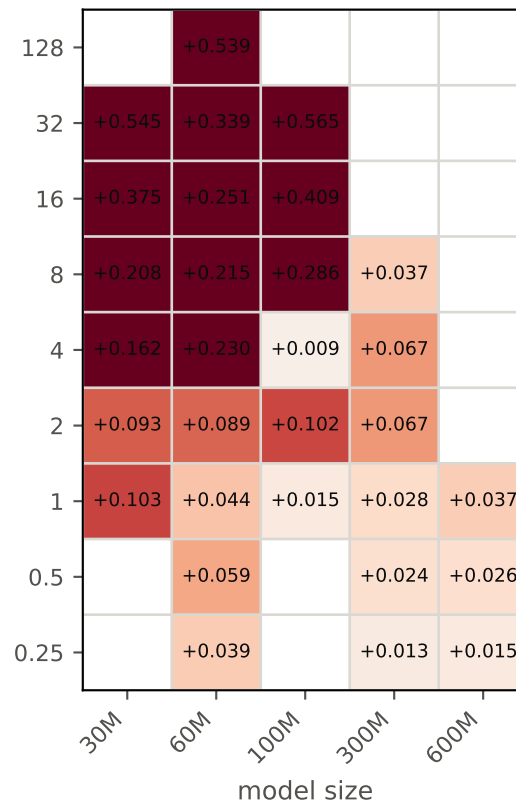
$\gamma = 0.005$



$\gamma = 0.01$



$\gamma = 0.02$



muon - adamw degradation, % of own unperturbed loss

cell = 100 x (perturbed - unperturbed) / unperturbed for muon minus the same for adamw | negative (blue) = muon degrades less | white = no data

colour scale clipped at ± 0.153 (85th pct); 14 cell(s) beyond it keep their printed value